

Unsloth Studio & Local UI Setup

1. System Overview & Architecture

Unsloth Studio provides a standalone local web user interface (UI) and orchestration CLI for training, fine-tuning, and deploying large language models (LLMs) on consumer hardware (Windows, Linux, WSL, and macOS) without requiring cloud GPU clusters.

- **Local Host Default:** `http://127.0.0.1:8888`
 - **Network Interface:** Pass `-H 0.0.0.0` to allow external device/LAN connections (e.g. tablet or phone access).
 - **Secure Remote Tunneling:** Launch via `unsloth studio --secure` to generate an encrypted Cloudflare HTTPS tunnel with zero-config port forwarding.
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2. Installation Runbook

macOS, Linux & WSL

Automated install script (configures isolated environment and packages)

```
curl -fsSL https://unsloth.ai/install.sh | sh
```

Launch Unsloth Studio on local port 8888

```
unsloth studio -p 8888 -H 0.0.0.0
```

Windows (PowerShell)

```
irm https://unsloth.ai/install.ps1 | iex
```

Isolated Developer / Staging Location

To prevent environment collisions with other agent runtimes, set `UNSLOTH_STUDIO_HOME`:

```
UNSLOTH_STUDIO_HOME="$HOME/.unsloth_studio" ./install.sh --local
```

```
UNSLOTH_STUDIO_HOME="$HOME/.unsloth_studio" unsloth studio -p 8888
```

3. Unsloth Start: Agent-to-Model Linking

Unsloth provides the `unsloth start` command to bind local models directly to command-line AI agents without manual reverse-proxy setup:

Connect local model to Claude Code CLI

```
unsloth start claude
```

Connect local model to Hermes 3 Agent

```
unsloth start hermes
```

Connect to OpenAI Codex / OpenClaw

```
unsloth start codex
```

```
unsloth start openclaw
```

4. Visual Data Recipes

Unsloth Studio includes a node-based visual workflow editor for data transformation:

1. Ingest raw PDFs, CSVs, and markdown documentation.
2. Clean, filter, and extract structured QA datasets.
3. Export directly to pre-tokenized training buffers for QLoRA and GRPO reinforcement learning.